

Finite element model updating of building structures under seismic excitation: A parallelized latent space–based Bayesian framework

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Abstract

Enhancing the probabilistic risk assessment of nuclear power plants requires accurate response predictions of reactor buildings under seismic hazards, typically obtained through dynamic analyses of high-fidelity finite element (FE) models. This paper presents a machine learning-aided Bayesian FE model updating framework for reinforced concrete (RC) building structures subjected to seismic excitation. The proposed framework employs a latent space-based likelihood approximation that transforms high-dimensional response data (e.g., time- or frequency-domain series) into low-dimensional embeddings using a Multimodal Variational Autoencoder (MVAE), enabling efficient evaluation of otherwise expensive likelihood functions. Experimental validation using shaking table test data from a six-story RC building specimen demonstrates that the proposed approach can estimate the Young's modulus with reasonably quantified uncertainty bounds.

Keywords: Finite element model updating, Approximate Bayesian computation, Deep generative model, Variational autoencoder, Latent space

1. Introduction

Enhancing the probabilistic risk assessment of nuclear power plants requires detailed response predictions of reactor buildings exposed to hazard-induced disturbances. Especially when addressing seismic hazards, dynamic analysis of high-fidelity finite element (FE) models plays a critical role (Nakamura et al., 2010; Choi et al., 2022). However, non-negligible discrepancies often exist between model predictions and the observed behavior of actual buildings due to various factors, e.g., measurement noise, variability in material properties, uncertainties introduced during construction process, over-simplification in the modeling of joints and boundaries, material degradation, and hazard-induced structural damage (Beck and Katafygiotis, 1998; Sinha and Friswell, 2003; Yaoyama et al., 2024). To address this, Bayesian FE model updating (Simoen et al., 2015; Huang et al., 2019; Kiran et al., 2025) provides a rigorous framework for updating model parameters with quantified uncertainties, thereby allowing for robust detection of structural damage and material degradation, enhanced response predictions for future seismic hazards, and risk-informed maintenance activities. In the field of nuclear engineering, several studies have adopted Bayesian methods for updating FE models of, e.g., containment vessels (Song et al., 2024) and reinforced concrete (RC) building structures (Meng et al., 2025).

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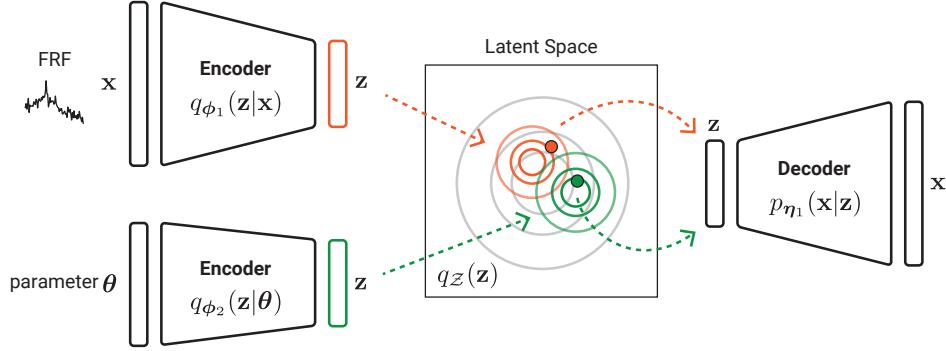


Figure 1: Schematic of Multimodal VAE architecture adopted in this study. **Notation 整理!**

fig:mvae

When considering high-fidelity FE models exposed to dynamic loading, the following challenges should be addressed. First, model responses are typically observed in the form of time- or frequency-domain series, representing high-dimensional data sequence with highly correlated dimensions. Explicitly defining likelihood functions for such sequential data presents a significant challenge (Lee et al., 2024, 2025). Additionally, evaluating these likelihood functions involves calls to an expensive simulator (e.g., an ODE solver), making posterior inference computationally demanding.

To address these challenges, machine learning-aided approximate Bayesian approaches have gained increasing attention (Zeng et al., 2025a,b; Wang et al., 2025; Wang and Bi, 2026). In such approaches, neural networks are trained in advance on parameter–response pairs generated by a simulator and are then used to approximate either the likelihood function or the posterior distribution. This framework avoids explicit likelihood evaluations and can substantially reduce the number of simulator calls during inference. Specifically, applications based on normalizing flows (Zeng et al., 2025a,b; Wang et al., 2025) and diffusion models (Wang and Bi, 2026) have recently been reported. Along the same line, our research group has developed a latent space-based Bayesian inference (LSBI) framework (Lee et al., 2024; Itoi et al., 2024; Lee et al., 2025). In LSBI, an approximate likelihood is formulated in a low-dimensional latent space, which circumvents direct likelihood modeling in the original high-dimensional observation space. When a multimodal variational autoencoder (MVAE) is used as the dimensionality reduction model, posterior inference can be performed without any additional simulator evaluations once the surrogate has been trained, resulting in an efficient amortized inference scheme (Itoi et al., 2024).

In this study, we examine the applicability of the latent space-based approach to high-fidelity FE models of building structures. We demonstrate the proposed approach using shaking table test data from a six-story reinforced concrete (RC) wall-frame structure (Sugimoto et al., 2017), targeting the Young’s modulus of concrete for each story and damping ratios as parameters of interest. Additionally, we present a comparative study with the method that explicitly assumes the likelihood function.

2. Latent space–based likelihood approximation revisited

Consider an engineering system characterized by uncertain parameters $\theta \in \Theta \subseteq \mathbb{R}^D$, with responses of interest denoted by $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^{n_x}$. We assume that the relationship between θ and $\mathbf{x}$ is described by a simulator, which induces a (possibly implicit) conditional distribution $p(\mathbf{x} | \theta)$. In structural engineering applications, the simulator typically corresponds to a deterministic finite element (FE) analysis. The data-

generating process is thus modeled as

$$\mathbf{x} = h(\boldsymbol{\theta}) + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon}), \quad (1)$$

where $h(\cdot)$ denotes the FE simulator and $\boldsymbol{\epsilon}$ represents measurement noise.

For simplicity of exposition, we consider the case where only a single observation $\mathbf{x}_{\text{obs}} \in \mathcal{X}$ is available. The extension to multiple observations is straightforward under the assumption of statistical independence. Using the Bayes' theorem, the posterior distribution of $\boldsymbol{\theta}$ conditioned on $\mathbf{x}_{\text{obs}}$ is given by

$$p(\boldsymbol{\theta} \mid \mathbf{x}_{\text{obs}}) \propto L(\boldsymbol{\theta}; \mathbf{x}_{\text{obs}}) p(\boldsymbol{\theta}), \quad (2)$$

where $L(\boldsymbol{\theta}; \mathbf{x}_{\text{obs}}) := p(\mathbf{x}_{\text{obs}} \mid \boldsymbol{\theta})$ denotes the likelihood function, and $p(\boldsymbol{\theta})$ and $p(\boldsymbol{\theta} \mid \mathbf{x}_{\text{obs}})$ denote the prior and posterior probability density functions, respectively. In engineering practice, such simulator-based inference often involves intractable or computationally prohibitive likelihood evaluations, due to the high dimensionality of observed responses (e.g., seismic response histories) and the cost of repeated FE simulations.

To address this issue, we introduce a lower-dimensional latent variable $\mathbf{z} \in \mathcal{Z} \subset \mathbb{R}^{n_z}$ such that $\mathbf{x} \perp \boldsymbol{\theta} \mid \mathbf{z}$ and $n_z \ll n_x$, leading to *latent space-based likelihood approximation* (Lee et al., 2024; Itoi et al., 2024):

$$\begin{aligned} L(\boldsymbol{\theta}; \mathbf{x}_{\text{obs}}) &= \int_{\mathcal{Z}} p(\mathbf{x}_{\text{obs}} \mid \mathbf{z}) p(\mathbf{z} \mid \boldsymbol{\theta}) d\mathbf{z} \\ &= c^{-1} \int_{\mathcal{Z}} \frac{p(\mathbf{z} \mid \mathbf{x}_{\text{obs}}) p(\mathbf{z} \mid \boldsymbol{\theta})}{p(\mathbf{z})} d\mathbf{z} \\ &\simeq c^{-1} \int_{\mathcal{Z}} \frac{q_{\phi_x}(\mathbf{z} \mid \mathbf{x}_{\text{obs}}) q_{\phi_\theta}(\mathbf{z} \mid \boldsymbol{\theta})}{p(\mathbf{z})} d\mathbf{z} \end{aligned} \quad \text{eq:latent} \quad (3)$$

where $c = p(\mathbf{x}_{\text{obs}}) = \int_{\boldsymbol{\theta}} p(\mathbf{x}_{\text{obs}} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}$ denotes a normalizing constant; $p(\mathbf{z})$ represents a prior distribution of $\mathbf{z}$ and is typically defined as a standard normal distribution, i.e., $p(\mathbf{z}) = \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I}_{n_z})$, where $\mathbf{I}_n$ denotes an $n \times n$ identity matrix. The densities q_{ϕ_x} and q_{ϕ_θ} represent probabilistic encoders parameterized ϕ_x and ϕ_θ , which provide shared embeddings from observed (or simulated) responses $\mathbf{x}$ and parameters $\boldsymbol{\theta}$. This formulation avoids the need for explicit likelihood evaluation in the high-dimensional observation space.

Following Itoi et al. (2024), we specifically adopt the probabilistic encoders from multimodal variational autoencoder (MVAE), which leads to the MVAE-based LSBI framework. MVAE is a neural network architecture designed to learn a shared latent representation across multiple data modalities (i.e., $\boldsymbol{\theta}$ and $\mathbf{x}$ in this study). As illustrated in Figure 1, the architecture employed here consists of one decoder $p_\eta(\mathbf{x} \mid \mathbf{z})$ and two encoders $q_{\phi_x}(\mathbf{z} \mid \mathbf{x})$ and $q_{\phi_\theta}(\mathbf{z} \mid \boldsymbol{\theta})$. The decoder p_η represents a probabilistic generative process from a latent variable $\mathbf{z} \sim p(\mathbf{z}) := \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I}_{n_z})$ to the response $\mathbf{x}$, and is modeled in a Gaussian form,

$$p_\eta(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_\eta(\mathbf{z}), \text{diag}(\boldsymbol{\sigma}_\eta^2(\mathbf{z}))), \quad \text{eq:peta} \quad (4)$$

where $\boldsymbol{\mu}_\eta(\cdot)$ and $\boldsymbol{\sigma}_\eta^2(\cdot)$ represent neural networks parameterized by η , and $\text{diag}(\cdot)$ denotes a diagonal matrix constructed from a given vector. The encoders q_{ϕ_x} and q_{ϕ_θ} have the following forms,

$$q_{\phi_x}(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_{\phi_x}(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_{\phi_x}^2(\mathbf{x}))), \quad \text{eq:qphi1} \quad (5)$$

$$q_{\phi_\theta}(\mathbf{z} \mid \boldsymbol{\theta}) = \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_{\phi_\theta}(\boldsymbol{\theta}), \text{diag}(\boldsymbol{\sigma}_{\phi_\theta}^2(\boldsymbol{\theta}))). \quad \text{eq:qphi2} \quad (6)$$

77 where $\mu_{\phi_{\mathbf{x}}}, \sigma_{\phi_{\mathbf{x}}}^2$ and $\mu_{\phi_{\theta}}, \sigma_{\phi_{\theta}}^2$ are neural networks parameterized by weights $\phi_{\mathbf{x}}$ and ϕ_{θ} , respectively. We
 78 accordingly define the MVAE-based likelihood approximation as

$$\hat{L}_{\phi_{\mathbf{x}}, \phi_{\theta}}(\theta; \mathbf{x}) := \int_{\mathcal{Z}} \frac{q_{\phi_{\mathbf{x}}}(\mathbf{z} | \mathbf{x}_{\text{obs}}) q_{\phi_{\theta}}(\mathbf{z} | \theta)}{p(\mathbf{z})} d\mathbf{z}, \quad (7)$$

79 Notably, as we adopt Gaussian forms for the encoders $q_{\phi_{\mathbf{x}}}(\mathbf{z} | \mathbf{x})$, $q_{\phi_{\theta}}(\mathbf{z} | \theta)$ and prior $p(\mathbf{z})$, the integral in
 80 Equation (3) reduces to a closed form (Lee et al., 2024) and can readily be evaluated.

81 In the present paper, we construct the approximate likelihood $\hat{L}_{\phi_{\mathbf{x}}, \phi_{\theta}}(\theta; \mathbf{x}_{\text{obs}})$ in an amortized manner
 82 as follows. First, we draw parameter samples from a predefined proposal distribution $\pi(\theta)$ (typically set to
 83 the prior $p(\theta)$), i.e., $\theta^{(n)} \sim \pi(\theta)$ for $n = 1, \dots, N_{\text{train}}$. For each $\theta^{(n)}$, we call a simulator $p(\mathbf{x} | \theta)$ to sample
 84 two independent samples $(\mathbf{x}^{(n)}, \tilde{\mathbf{x}}^{(n)})$. In the case of a deterministic simulator with additive observation
 85 noise, this reduces to

$$\mathbf{x}^{(n)} = h(\theta^{(n)}), \quad (8)$$

$$\tilde{\mathbf{x}}^{(n)} = h(\theta^{(n)}) + \epsilon^{(n)}, \quad \epsilon^{(n)} \sim p(\epsilon), \quad (9)$$

86 where the noise model $p(\epsilon)$ is specified based on prior knowledge or engineering expertise. Using the
 87 resulting dataset $\mathcal{D} = \{(\theta^{(n)}, \mathbf{x}^{(n)}, \tilde{\mathbf{x}}^{(n)})\}$, we train neural networks $(\eta, \phi_{\mathbf{x}}, \phi_{\theta})$ by minimizing the following
 88 loss function,

$$\mathcal{L}(\eta, \phi_{\mathbf{x}}, \phi_{\theta}) = \mathcal{L}_{\text{recon}}(\eta, \phi_{\mathbf{x}}) + \mathcal{L}_{\text{pred}}(\eta, \phi_{\theta}) + \mathcal{L}_{\text{regul}}(\phi_{\mathbf{x}}, \phi_{\theta}), \quad \text{eq:loss} \quad (10)$$

89 where the reconstruction, prediction, and regularization terms are defined as

$$\mathcal{L}_{\text{recon}}(\eta, \phi_{\mathbf{x}}) = -\mathbb{E}_{q_{\phi_{\mathbf{x}}}(\mathbf{z} | \mathbf{x})} [\log p_{\eta}(\tilde{\mathbf{x}} | \mathbf{z})], \quad (11)$$

$$\mathcal{L}_{\text{pred}}(\eta, \phi_{\theta}) = -\mathbb{E}_{q_{\phi_{\theta}}(\mathbf{z} | \theta)} [\log p_{\eta}(\tilde{\mathbf{x}} | \mathbf{z})], \quad (12)$$

$$\begin{aligned} \mathcal{L}_{\text{regul}}(\phi_{\mathbf{x}}, \phi_{\theta}) = & D_{\text{KL}}(q_{\phi_{\mathbf{x}}}(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z})) + D_{\text{KL}}(q_{\phi_{\theta}}(\mathbf{z} | \theta) \| p(\mathbf{z})) \\ & + \alpha \left[D_{\text{KL}}(q_{\phi_{\mathbf{x}}}(\mathbf{z} | \mathbf{x}) \| q_{\phi_{\theta}}(\mathbf{z} | \theta)) + D_{\text{KL}}(q_{\phi_{\theta}}(\mathbf{z} | \theta) \| q_{\phi_{\mathbf{x}}}(\mathbf{z} | \mathbf{x})) \right]. \end{aligned} \quad (13)$$

90 Here, $\mathbb{E}_p[\cdot]$ denotes the expectation with respect to a probability distribution p , and $D_{\text{KL}}(p \| q)$ denotes the
 91 Kullback–Leibler divergence between distributions p and q . The hyperparameter α controls the strength of
 92 the consistency regularization between the two encoders. In this study, we set $\alpha = 5$. By defining $\mathcal{L}_{\text{recon}}$
 93 and $\mathcal{L}_{\text{pred}}$ with respect to independent response samples (or the deterministic and contaminated responses),
 94 the MVAE is trained to reflect stochastic data-generating processes.

95 各損失の位置付けについても少し触れる？

96 3. Parallelized LSBI framework with Sequential Monte Carlo sampler

97 3.1. Sequential Monte Carlo

98 To infer the (approximated) posterior $\hat{p}(\theta | \mathbf{x}_{\text{obs}}) \propto \hat{L}_{\phi_{\mathbf{x}}, \phi_{\theta}}(\theta; \mathbf{x}_{\text{obs}}) p(\theta)$, we adopt SMC sampler
 99 (Ching and Chen, 2007; Betz et al., 2016; Carrera and Papaioannou, 2024). The general algorithm of SMC
 100 is formulated as follows. We initially place a population of samples (particles) drawn from the prior $p(\theta)$,
 101 and gradually push them to the target posterior via the suite of *intermediate* PDFs,

$$p_t(\theta) = c^{-1} \{ \hat{L}_{\phi_{\mathbf{x}}, \phi_{\theta}}(\theta; \mathbf{x}_{\text{obs}}) \}^{\beta_t - \beta_{t-1}} p_{t-1}(\theta), \quad \text{for } t = 1, \dots, T. \quad (14)$$

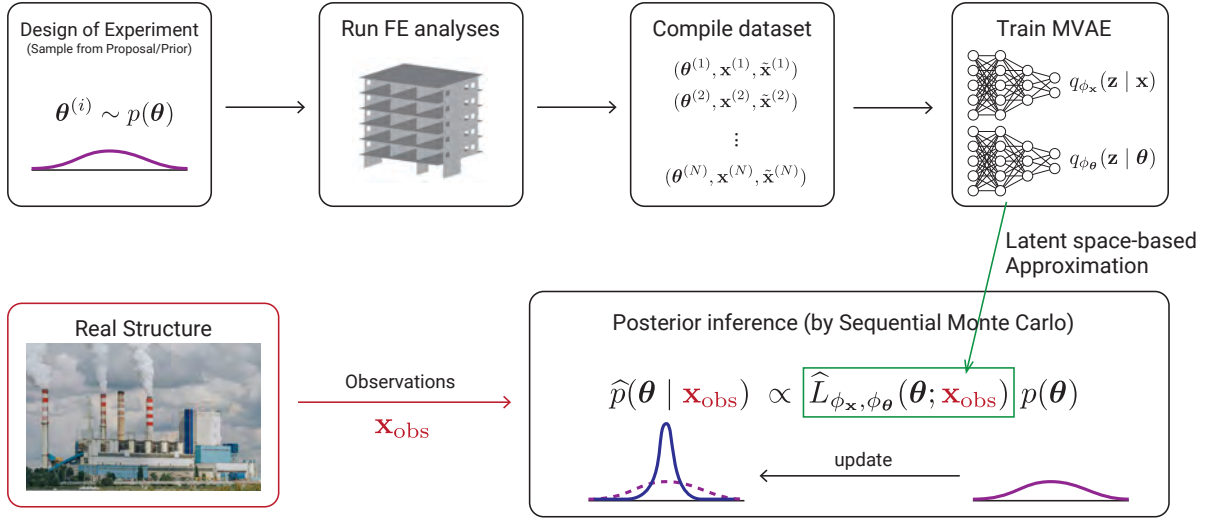


Figure 2: The overview of the proposed framework: LSBI-SMC.

fig:flowchart

102 $p_0(\theta) = p(\theta)$ and the resulting distribution $p_T(\theta)$ corresponds to $\hat{p}(\theta | \mathbf{x}_{\text{obs}})$. β_t is a tempering parameter
 103 that starts from $\beta_0 = 0$ and monotonically increases until $\beta_T = 1$. At each level t , β_t is determined based
 104 on the particles $\{\theta_{t-1}^{(n)}\}_{n=1}^N$ at the previous level by the following rule.

$$\beta_t = \operatorname{argmin}_{\beta \in (\beta_{t-1}, 1]} \left\{ N_{\text{eff}}(\beta; \{\theta_{t-1}^{(n)}\}_{n=1}^N) - \gamma N \right\} \quad (15)$$

105 where N_{eff} denotes the effective sample size (ESS) described as

$$N_{\text{eff}}(\beta; \{\theta_{t-1}^{(n)}\}_{n=1}^N) = \frac{\left(\sum_{n=1}^N w_t^{(n)}(\beta) \right)^2}{\sum_{n=1}^N w_t^{(n)}(\beta)^2} \quad (16)$$

106 with $w_t^{(n)}(\beta) = L(\theta_{t-1}^{(n)} | \mathbf{x}_{\text{obs}})^{\beta - \beta_t}$ denoting the weight of each particle for a specified β . The hyperpa-
 107 rameter γ controls the degree of transition with γN representing the target ESS.

108 The update of particles at each level t consists of two steps: “resample” and “move”. In the former
 109 step, N particles are resampled according to the weights $w_t^{(n)}(\beta_t)$. In the latter step, MCMC updates with
 110 a Markov transition kernel $\mathcal{K}(\theta, \cdot)$ are performed for each particle. We specifically adopt the random walk
 111 Metropolis–Hastings (RWMH) kernel, in which a candidate position is sampled according to $\theta^* \sim \mathcal{K}(\theta, \cdot)$
 112 and evaluated according to the acceptance probability given by

$$\eta = \min \left\{ 1, \frac{\mathcal{K}(\theta^*, \theta) p_t(\theta^*)}{\mathcal{K}(\theta, \theta^*) p_t(\theta)} \right\}. \quad (17)$$

113 The transition kernel is typically taken as a symmetric distribution such as Gaussian, $\mathcal{K}(\theta, \cdot) = \mathcal{N}(\cdot | \theta, \Sigma)$,
 114 resulting in the acceptance probability with a reduced form, $\eta = \min\{1, p_t(\theta^*)/p_t(\theta)\}$. Ching and Chen
 115 (2007) proposed the use of a Gaussian proposal with a covariance,

$$\Sigma_t = b^2 \cdot \sum_{n=1}^N \frac{w_t^{(n)}}{S_t \cdot N} (\theta_t^{(n)} - \bar{\theta}_t)(\theta_t^{(n)} - \bar{\theta}_t)^\top \quad \text{eq:ching_proposal} \quad (18)$$

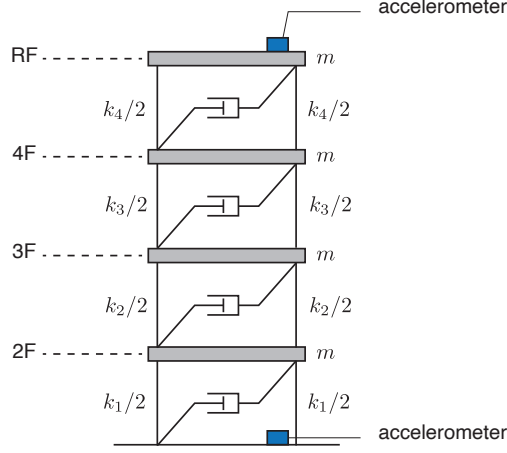


Figure 3: The target structure in the benchmarking example: a shear-type 4DOF building model. `fig:shear4dof`

where $S_t = \sum_{n=1}^N w_t^{(n)} / N$ and $\bar{\theta}_t = \sum_n w_t^{(n)} \theta_t^{(n)} / \sum_n w_t^{(n)}$. In the present paper, we set hyperparameters as $\gamma = 0.8$ and $b = 0.2$.

3.2. Proposed framework

Our proposed framework is summarized as follows.

1. Draw parameter samples from a predefined proposal distribution $\pi(\theta)$ (typically set to the prior distribution $p(\theta)$): $\theta^{(n)} \sim \pi(\theta)$ for $n = 1, \dots, N_{\text{train}}$.
2. For each $\theta^{(n)}$, generate two independent response samples from the simulator: $\mathbf{x}^{(n)}, \tilde{\mathbf{x}}^{(n)} \sim p(\mathbf{x} | \theta^{(n)})$, resulting in a training dataset $\mathcal{D} = \{(\theta^{(n)}, \mathbf{x}^{(n)}, \tilde{\mathbf{x}}^{(n)})\}_{n=1}^{N_{\text{train}}}$.
3. Using $\mathcal{D}$, train the neural networks $(\eta, \phi_{\mathbf{x}}, \phi_{\theta})$ by minimizing the loss function $\mathcal{L}$ in Equation (10) to construct the approximate likelihood function $\hat{L}_{\phi_{\mathbf{x}}, \phi_{\theta}}(\theta; \mathbf{x})$.
4. For a given observation $\mathbf{x}_{\text{obs}}$, run the SMC sampler to draw posterior samples from $\hat{p}(\theta | \mathbf{x}_{\text{obs}}) \propto \hat{L}_{\phi_{\mathbf{x}}, \phi_{\theta}}(\theta; \mathbf{x}_{\text{obs}}) p(\theta)$.

The above procedure is illustrated in Figure 2.

なぜこの手法が良いのか、という点をあらためてまとめる。 amortized, GPU-enhanced, ...

4. Benchmarking example

4.1. Target structure and data

The target structure is a shear-type 4DOF building structure as illustrated in Figure 3. We assume that the masses for all floors have an identical value of m (kg). Rayleigh damping is adopted with damping ratio of 0.02 both at 1.0 and 30.0 Hz. The stiffness values of all stories are parameterized as $k_i = \theta_i \bar{k}$ with $\bar{k} = 1000 \cdot m \text{ (s}^{-2}\text{)}$ denoting a representative value. The coefficients $\theta := \{\theta_i\}_{i=1}^4$ are uncertain parameters to be updated with uniform priors $\theta_i \sim \mathcal{U}(\cdot | [0.33, 3.00])$ ($i = 1, \dots, 4$), where $\mathcal{U}(\cdot | S)$ denotes the pdf of a uniform distribution defined over the set S . In this example, we assume that only base and roof top acceleration measurements are available. Under this assumption, the parameter θ is **not globally identifiable** and permit several equivalent solutions (Katafygiotis and Beck, 1998). We set $\theta = \{1, 1, 1, 1\}$ as ground

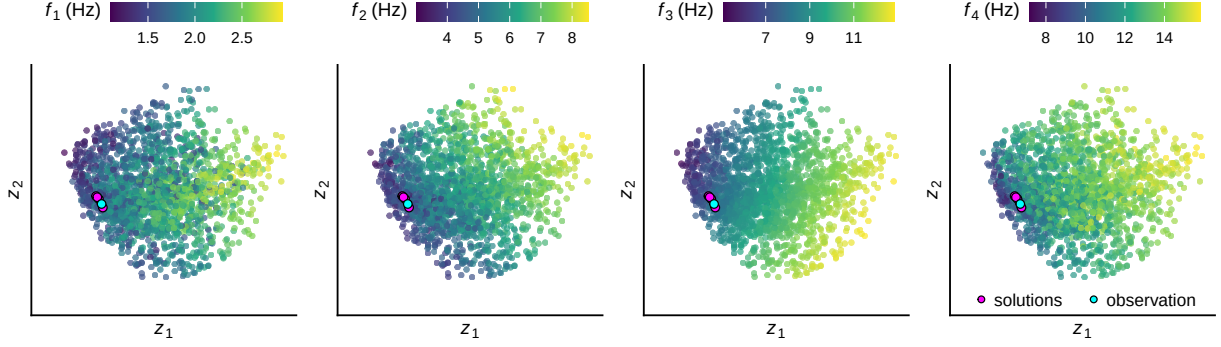


Figure 4: Visualization of the latent space of the trained MVAE model.

fig:latent

truth values and apply the algorithm presented in Katafygiotis and Beck (1998) to identify the other three solutions that reproduce the identical roof-top response, as listed in Table 1. This implies that our target posterior exhibits severe multimodality.

The quantities of interest herein are the frequency response function (FRF) of roof-top acceleration with respect to base excitation. We define a simulator as $h : \theta \mapsto \mathbf{x}$, where $\mathbf{x} \in \mathbb{R}^{1024}$ represents the logarithm of the absolute FRF, $\log |H(f)|$, evaluated at the frequency points $f_n = \{0.00, 0.02, \dots, 20.46\}$ Hz with additive observation noise. This reads

$$x_n = \log |H(f_n)| + \epsilon_n, \quad \epsilon_n \sim \mathcal{N}(0, \sigma_\epsilon^2), \quad (19)$$

for $n = 1, \dots, 1024$. We set $\sigma_\epsilon = 0.2$.

4.2. Training of MVAE

We draw 10^5 samples of $\theta^{(n)}$ from the prior and run the simulator to acquire responses $\mathbf{x}^{(n)} = h(\theta^{(n)})$, yielding the training dataset $\{(\theta^{(n)}, \mathbf{x}^{(n)})\}_{n=1}^{10^5}$. This is randomly split into training and validation sets, comprising 90% and 10% of the samples, respectively. We train an MVAE architecture that consists of three neural networks, i.e., two probabilistic encoders q_{ϕ_1} and q_{ϕ_2} as well as a single probabilistic decoder p_η . All neural networks include fully connected (FC) layers and residual blocks composed of convolutional layers. The dimension of latent space is set to $n_z = 8$. For further details in the architecture, we refer the readers to our GitHub repository, To optimize these networks, we adopt the Adam algorithm with a learning rate of 10^{-3} and batch size of 256. The hyperparameter in the loss function in Eq. (??) is set to $\alpha = \dots$. The number of training epochs are determined using the early stopping strategy; that is, the training is stopped when the validation loss does not improve for 20 epochs. The whole training procedure is implemented using a Python package PyTorch.

Table 1: Equivalent parameters that reproduce the identical roof response in the 4DOF example.

Number#	k_1	k_2	k_3	k_4
1 (ground truth)	1.000	1.000	1.000	1.000
2	1.722	0.636	1.301	0.701
3	1.999	1.000	0.500	1.000
4	2.640	0.647	0.813	0.720

Table 2: Results in the 4DOF benchmark example for each sampling algorithm (summary of independent 10 runs).

tab:results					
Algorithm	N_s	N_{eval}	MMD		Wall-clock time (sec) [†]
			mean	std.	
SMC	500	91000	0.086	0.038	0.8
	1000	182000	0.077	0.034	0.8
	2000	362000	0.051	0.019	0.8
	4000	724000	0.040	0.015	0.8
NUTS	—	558871	0.629	0.159	1782

[†] These do not include the elapsed time required for dataset construction and MVAE training.

We investigate the geometric structure of the latent space learned by the trained MVAE. For illustration, we draw prior samples $\theta^{(n)} \sim p(\theta)$ and map them to the latent space through the encoder $q_{\phi_2}(\mathbf{z} \mid \theta)$. We then apply principal component analysis (PCA) to the mean latent vectors $\mu_{\mathbf{z}} \in \mathbb{R}^8$ for visualization. The resulting two-dimensional projections are shown in Figure 4. In each panel, color indicates the corresponding natural frequency f_i ($i = 1, \dots, 4$). The projections are organized according to the natural frequencies, implying that the learned latent space successfully captures the intrinsic features of the FRFs. We additionally annotate the four equivalent solutions listed in Table 1, mapped into the latent space via the encoder $q_{\phi_1}(\mathbf{z} \mid \theta)$, as well as the observation generated from the ground truth $\theta = \{1, 1, 1, 1\}^\top$, which is mapped through the encoder $q_{\phi_2}(\mathbf{z} \mid \mathbf{x})$. Although these equivalent solutions are widely separated in the original parameter space, all equivalent solutions are located in close proximity to the observation in the latent space. This indicates that the MVAE successfully retrieves the shared latent representation between the parameters θ and responses $\mathbf{x}$.

4.3. Posterior sampling

Using the MVAE-based likelihood approximation $L_{\text{MVAE}}(\theta; \mathbf{x})$, we draw the posterior samples from $p_{\text{post}}(\theta \mid \mathbf{x}) \propto L_{\text{MVAE}}(\theta; \mathbf{x}) p(\theta)$. We benchmark two sampling algorithms: SMC and No-U-Turn Sampler (NUTS) (Hoffman and Gelman, 2014). NUTS is the state-of-the-art MCMC algorithm and a variant of Hamiltonian Monte Carlo (HMC) (Duane et al., 1987; Neal, 2011). HMC (Duane et al., 1987; Neal, 2011) leverages Hamiltonian dynamics to construct gradient-informed proposals using a leapfrog integrator, leading to high acceptance rates and efficient exploration. NUTS adaptively determines the trajectory length, removing the need for manual tuning of integration parameters. Since these methods require gradients of the log-likelihood, we compute them efficiently using automatic differentiation.

In this benchmark test, we run a single NUTS chain for 600,000 iterations, discarding the first 10,000 iterations as burn-in. Note that the number of MCMC iterations does not coincide with the number of likelihood evaluations, since each iteration includes multiple leapfrog steps, each of which requires evaluations of the likelihood function and its gradient. The implementation is based on the Python package Pyro. For SMC, we consider three population sizes $N_s \in \{1000, 2000, 4000\}$, while fixing the number of MCMC iterations per round to $N_{\text{iter}} = 10$. These configurations are chosen such that total number of likelihood evaluations (denoted by N_{eval}) of L_{MVAE} is comparable between the SMC and NUTS algorithms.

To quantify the performance of the sampling algorithms, we use the maximum mean discrepancy (MMD) Gretton et al. (2012), a kernel-based metric that measures the discrepancy between two sample distributions. As a reference, we generate posterior samples using SMC with a large population size of $N_s = 5 \times 10^4$. For each inference setting, we compute the MMD between the resulting posterior samples and this reference set; that is, a lower MMD value indicates higher inference performance. In the MMD calculation, we only use 5000 samples randomly selected from the reference set due to computational efficiency.

Table 2 summarizes the performance for different inference settings averaged over independent ten runs, while the likelihood estimator L_{MVAE} is common across all runs and settings. SMC, for all considered

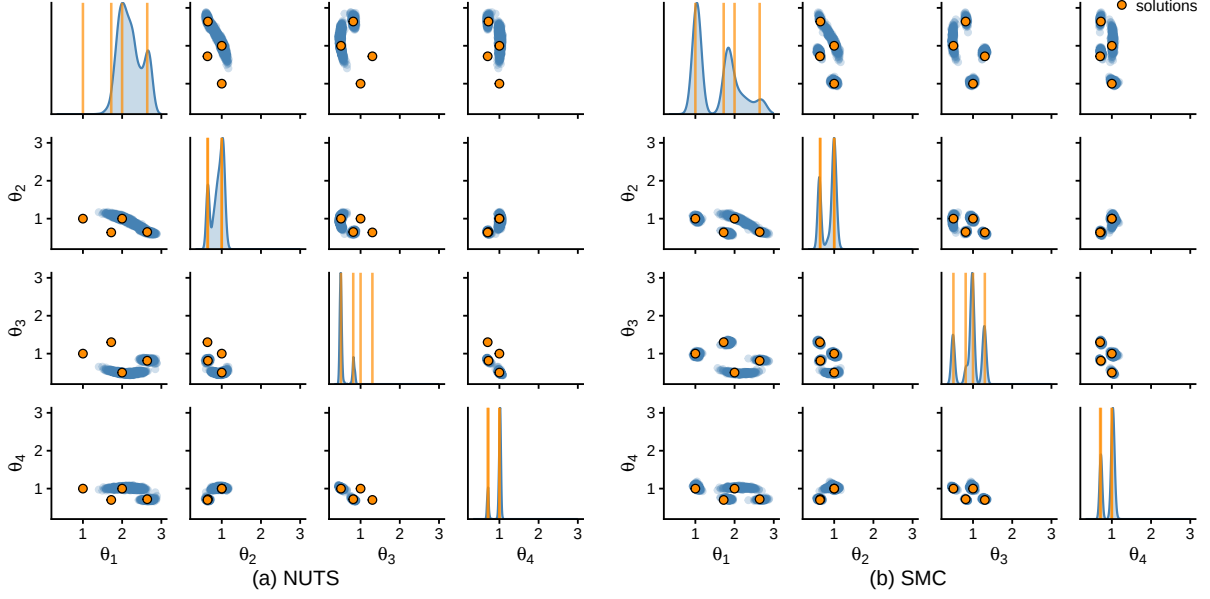


Figure 5: The distribution of posterior samples in the benchmarking example (SMC-HMC, $N_s = 2000$). fig:post_4dof

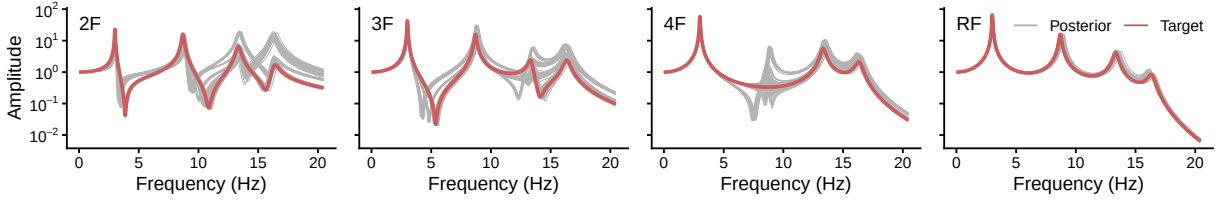


Figure 6: Posterior predictions of FRF in the benchmarking example (SMC-HMC, $N_s = 2000$). fig:pred_4dof

values of N_s , exhibits superior performance compared to NUTS in terms of MMD. Notably, SMC also requires substantially less wall-clock time than NUTS, highlighting the computational efficiency enabled by GPU-parallelized implementation.

Figure 5 displays the distributions of posterior samples obtained from NUTS and SMC ($N_s = 2000$). While NUTS fails to fully explore the four posterior modes associated with the equivalent solutions, SMC successfully captures all modes. Figure 6 presents the predictive distribution of FRFs on all floors, which are simulated from posterior samples in SMC ($N_s = 2000$), as well as the target FRF for the ground-truth parameter $\theta = \{1, 1, 1, 1\}^\top$. For the roof-top response (RF), the posterior predictive distributions closely align with the observed target, indicating that the inference successfully fits the measurement. For the remaining unobserved floors (2F, 3F, and 4F), the predicted FRFs exhibit clear multimodality while still covering the target responses. These results demonstrate that the proposed LSBI-SMC approach maintains robust exploration capability even in severely ill-posed inference problems characterized by multimodal posteriors. This highlights its suitability for model updating under practical sensing constraints.

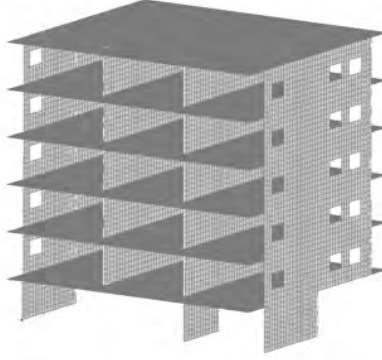


Figure 7: Finite element model of the target specimen. 耐震壁の位置などを描き込む.

fig:femodel

5. Application to RC building structures under seismic excitation

5.1. Target structure and data

The target structure is a scaled specimen of a six-story RC wall-frame building structure from the E-Defense shaking table test (Sugimoto et al., 2017). Figure 7 displays the developed FE model for the target specimen. The specimen has shear walls in the X2 and X3 frames and has non-structural walls in the X1 and X4 frames. We specifically focus on the loading test termed Case #1–2 in Sugimoto et al. (2017) under white noise excitation, and use acceleration measurements only in the X-direction. In this study, response time histories from multiple sensors installed on each floor are averaged in the time domain and transformed into the absolute value of FRFs with respect to base acceleration. Namely, we consider a $N_{\text{freq}} \times N_{\text{story}}$ data tensor for $\mathbf{x}_{\text{obs}}$, where $N_{\text{freq}} = 1024$ is the number of considered frequency points and $N_{\text{story}} = 6$ corresponds to all six floors (2F, 3F, ..., RF).

The developed FE model displayed in Figure 7 has shell elements for shear walls and slabs and beam elements for beams and columns. The fixed boundary conditions are introduced at the bottom surfaces of the columns and walls in the first story. FE analyses are conducted using a commercial software, TDAP III. We parameterize the Young's modulus of concrete for the i th story as $E_i = \theta_i E_0$, where $E_0 = 2.44 \times 10^4$ (N/mm²) denotes a nominal value, and θ_i represents a calibrating coefficient to be inferred. Poisson ratio is set as 0.2. The constant damping ratio ζ is assumed for all vibration modes, where ζ is also parameterized as $\zeta = \theta_7 \zeta_0$ with $\zeta_0 = 0.02$ denoting the nominal value. These configuration results in the parameter vector to be estimated as $\boldsymbol{\theta} = \{\theta_1, \dots, \theta_7\}^\top \in \mathbb{R}^7$.

To investigate the effect of sensor placement on the parameter estimation uncertainty, we compare two observation scenarios. Case 1 corresponds to the situation where the FRFs at all stories are observed, whereas Case 2 assumes that only the roof-top FRF is available. In Case 2, the parameter vector $\boldsymbol{\theta}$ is expected to be only partially identifiable due to the limited observation.

5.2. Training of MVAE

We adopt the same MVAE architecture as that described in 4.2. For the training dataset, we generated 100,000 samples of $\boldsymbol{\theta}$ from the uniform distribution $\boldsymbol{\theta} \sim \mathcal{U}(\cdot \mid [1/3, 3]^7)$. Linear dynamic FE analyses under white-noise excitation were conducted based on eigenvalue analysis to construct the synthetic dataset $\{(\boldsymbol{\theta}_i, \mathbf{x}_i)\}$. Ninety percent of the dataset is used for training, and the remaining ten percent for validation.

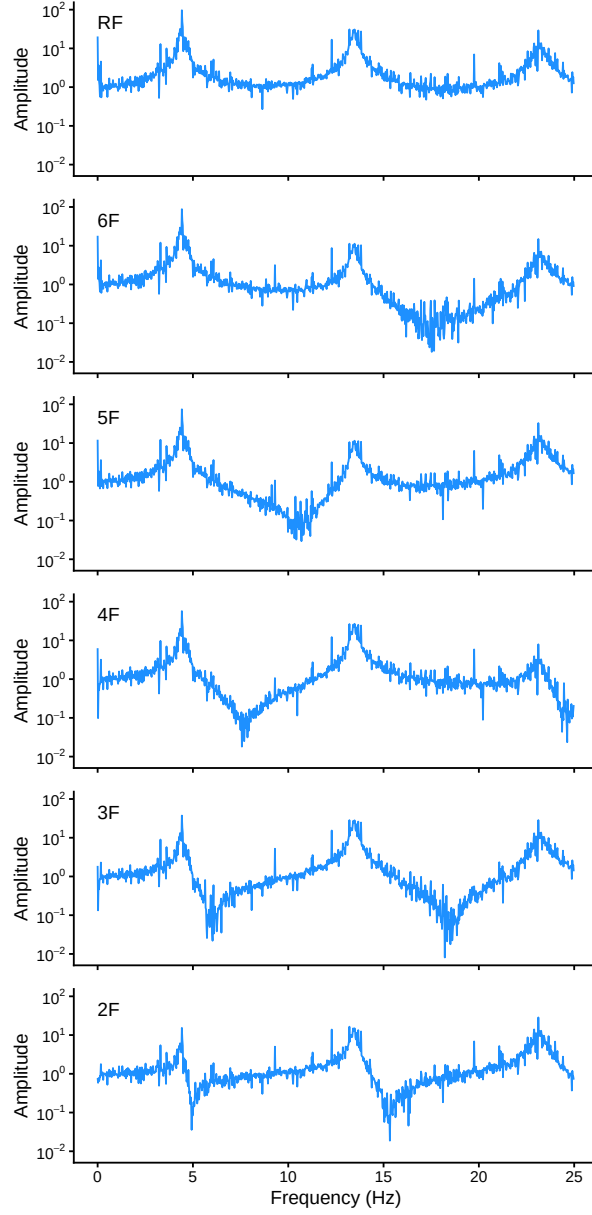


Figure 8: Observed frequency response functions in the experimental data (Sugimoto et al., 2017). `fig:exp-obs`

For training, θ_i is rescaled to the range $[0, 1]$, while $\mathbf{x}$ is standardized to have zero mean and unit variance, across the whole training dataset. The MVAE was trained using the Adam optimizer with a batch size of 256 and learning rate of 5×10^{-4} . Similarly to 4.2, the training is stopped with no improvement for 20 epochs.

5.3. Results

For both Cases 1 and 2, we perform LSBI-SMC with a population size of 50000 and hyperparameters set to $\gamma = 0.8$ and $b = 0.2$. The prior distributions for the Young's modulus calibration parameters are

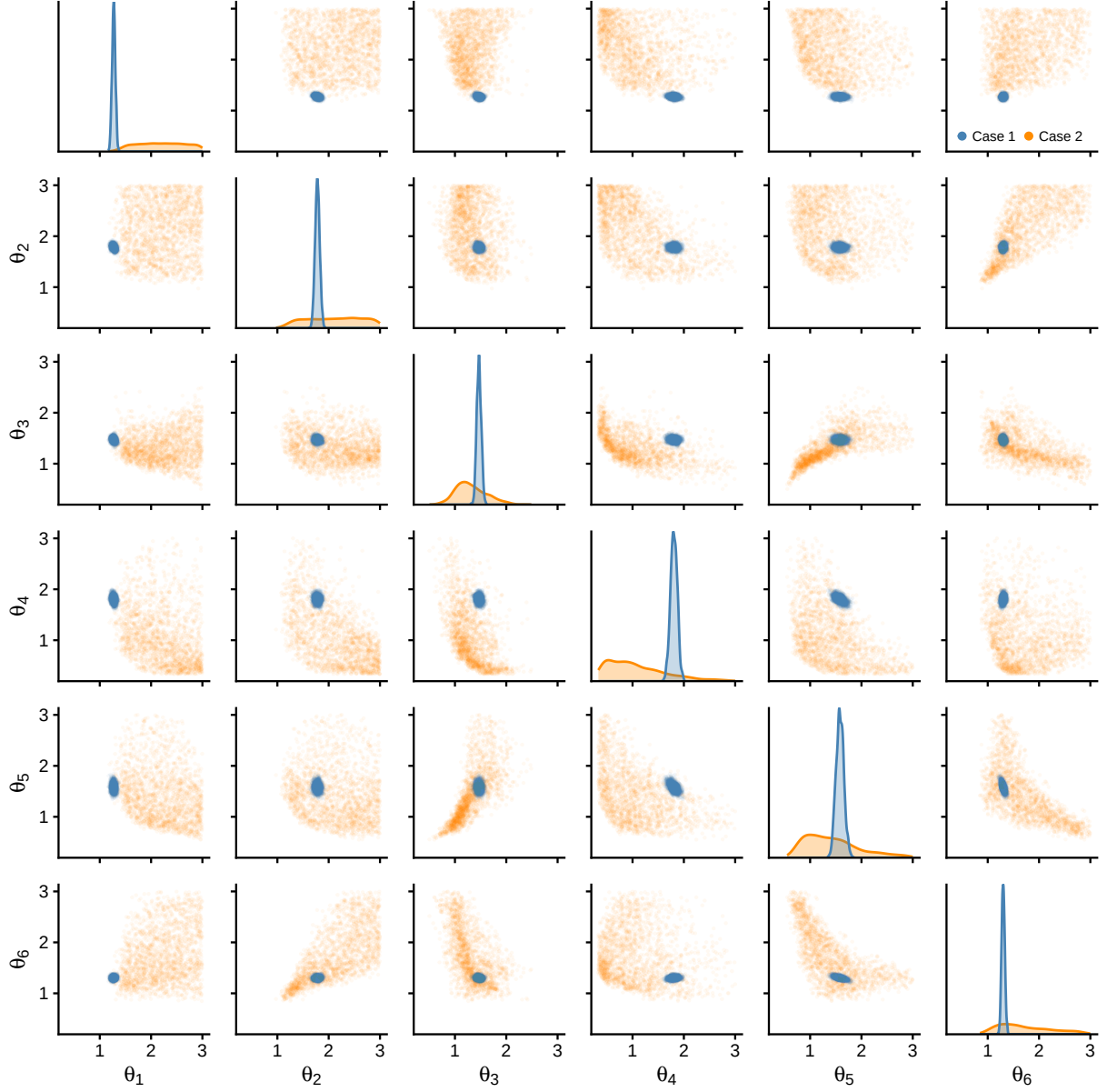


Figure 9: Inferred posterior samples for Case 1 (fully observed) and Case 2 (only the roof-top FRF observed). fig:exp-post

defined as $\theta_1, \dots, \theta_6 \sim \mathcal{U}(\cdot \mid [1/3, 3])$. For the damping ratio calibration parameter, a more informative prior is assumed as $\theta_7 \sim \mathcal{N}(\cdot \mid 1, 0.1^2)$, based on a pre-identified damping ratio obtained through system identification techniques, reflecting its weaker sensitivity to the FRFs.

Figure 9 presents a scatter plot matrix of 2,000 randomly selected posterior samples for $\theta_1, \dots, \theta_6$. The marginal posterior densities are shown along the diagonal. The posterior distribution in Case 2 exhibits a broader spread, reflecting the limited identifiability induced by sparse measurements. In contrast, the posterior distribution in Case 1 is contained within that of Case 2 but substantially narrower, suggesting that increasing the number of instrumented floors improves parameter identifiability and reduces the uncertainty

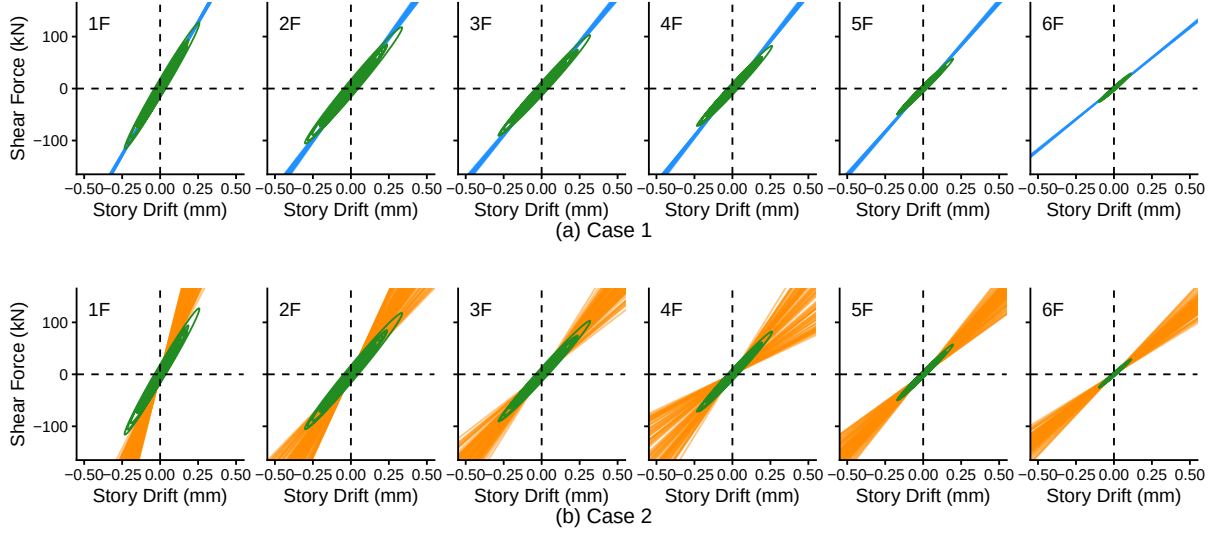


Figure 10: Predicted Q - d relationships for the 10%-scaled JMA Kobe excitation. 凡例

fig:exp-stif

in parameter estimation.

To validate the inferred posteriors, we predict seismic responses under a different excitation (namely the 10%-scaled JMA Kobe record), corresponding to Case #1–3 defined in Sugimoto et al. (2017). Figure 10 presents the predicted story stiffness obtained from static FE analyses based on 100 posterior samples. The predictions are compared with the observed Q - d relationships, where the story shear force Q (kN) is computed by multiplying the acceleration responses by the story mass, and the story drift d (mm) is measured using laser displacement sensors. In Case 1, the estimated story stiffness shows good agreement with the observed Q - d relationships for all stories. In Case 2, the predicted story stiffness exhibits a wider spread across all stories, reflecting increased uncertainty due to limited sensor placement. Nevertheless, the predicted Q - d relationships remain generally consistent with the observations for most stories, indicating that the proposed method captures the overall structural response even under reduced observability. For the first story, the predictions deviate from the observations, reflecting the discrepancy of the posterior distribution of θ_1 between Case 1 and Case 2, as shown in Figure 9. Figure 11 presents the predictive distributions of acceleration responses obtained from dynamic FE analyses using 100 posterior samples. The $\pm 2\sigma$ intervals, estimated from these samples, are shown together with the observed responses. For both cases, the predictions agree well with the observations, particularly around 11 s where large amplitudes are observed. Although Case 2 exhibits greater predictive uncertainty than Case 1, the observed responses are still well captured, demonstrating the robustness of the proposed model updating framework under sparse sensor placement.

6. Conclusions

This study proposed a machine learning-aided Bayesian FE model updating framework for RC building structures under seismic loading, utilizing a latent space inferred by a Multimodal Variational Autoencoder (MVAE). Experimental validation using shaking table test data demonstrated that the proposed method can effectively estimate the Young's modulus with reasonably quantified uncertainty bounds. The inferred posterior distributions were generally consistent with those obtained using an explicit likelihood formulation.

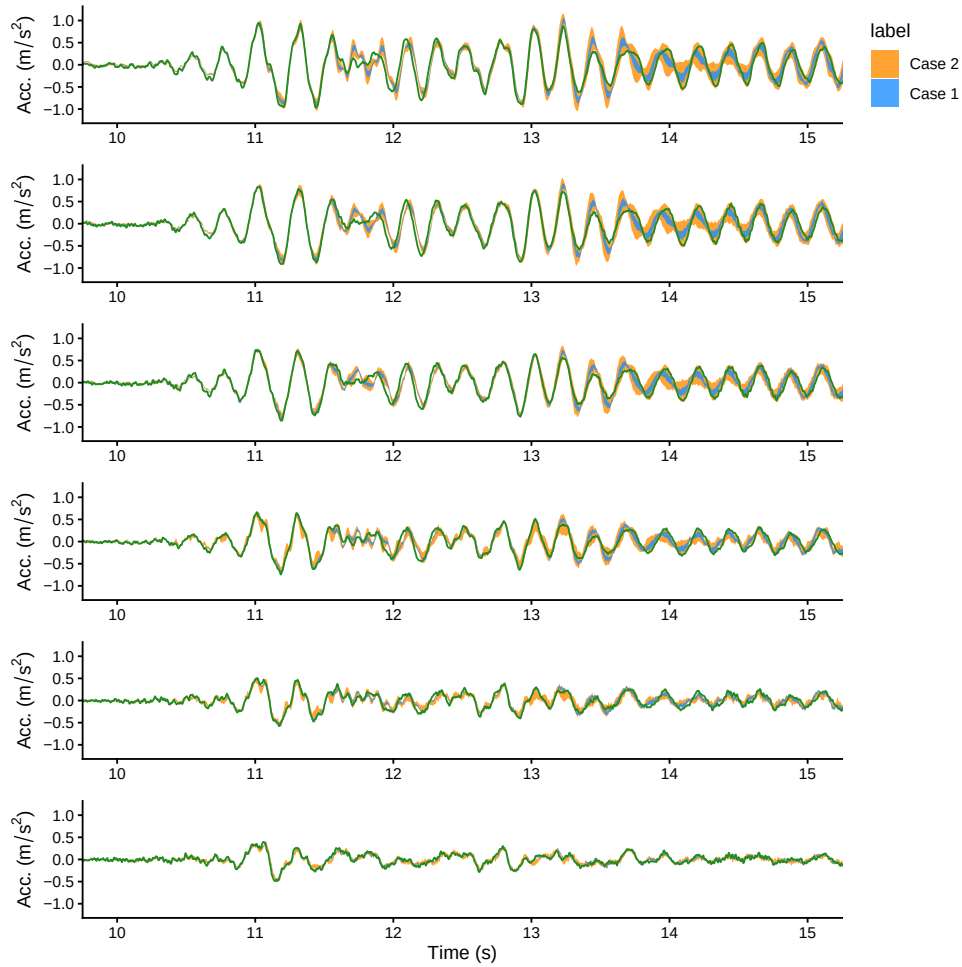


Figure 11: Posterior distributions of time history responses for the 10%-scaled JMA Kobe excitation. 凡例 fig:exp-pred

Moreover, the predictive distributions of story stiffness and acceleration responses showed good agreement with experimental observations, confirming the validity of the proposed approach.

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